

```

ds/realms/sandbox/">Compute Sandbox Cloud</a>
    </td>
    <td>
        This is for sandbox and ephemeral testing resources that provides an
        account/project for each user that are self-administered by each team member.
    </td>
</tr>
</tbody>
</table>

```

- The gitter realm was removed on 2021-07-22.

Department Realms

```

<table>
  <thead>
    <tr>
      <th style="width: 25%;">Value</th>
      <th style="width: 25%;">Human Friendly Name</th>
      <th style="width: 50%;">Description</th>
    </tr>
  </thead>
  <tbody>
    <tr>
      <td><code>eng-infra</code></td>
      <td>
        <a href="/handbook/company/infrastructure-standards/realms/eng-
infra/">Engineering Infrastructure</a>
      </td>
      <td>
        This is for additional services managed by Engineering Infrastructure and Site
        Reliability Engineers that may not be specific to GitLab.com SaaS (Ex. tools, release and
        package management services, etc).
      </td>
    </tr>
  </tbody>
</table>

```

- The sales-cs realm was migrated to the business-tech realm on 2021-07-22.

Environment Type (glenvtype)

```

{{% alert color="success" %}}
This label/tag is required.
{{% /alert %}}

```

```

terraform
glenvtype: 'experiment'

```

Expected values

Value	Description
experiment	(default) This is for individual user experiments are not impacted if powered off by infrastructure or security team, or data lost due to systems change.
test	This is for experiments that affect multiple users (ex. customer proof-of-concept or load testing) and would have disruptive impact if powered off.
stg	This is for a staging environment for tools/etc that would have major impact if powered off. These are for environments that are known to infra/security teams. The infra team has standardized on the use of stg instead of stag or stage, so we've universally adopted that here.
prd	This is for a production or production-like environment for tools/etc that would have major impact if powered off. These are for environments that are known to infra/security teams. The infra team has standardized on the use of prd instead of prod, so we've universally adopted that here.
other	As a last resort, you can use the other category. These are subject to periodic review.

> New values can be added to this list and used without any other changes needed. This list was created for this label/tag and is not inherited from another source.

Environment Name (glenvname)

```
{{% alert color="success" %}}
This label/tag is required.
{{% /alert %}}
```

```
terraform
glenvname: 'learning-kubernetes'
```

The environment name is a short name or description of the purpose of the environment or service. This is a sub-descriptor of grealm and glenvtype. The standards and use of this label/tag vary based on the grealm that it is used in, and each team can use it however it makes sense to them.

Realm Usage Guidelines

Realm	Usage Guidelines or Standards
infra-shared-services	(Placeholder for future use)
saas	(Placeholder for future use)
business-tech	(Placeholder for future use)
eng-dev	(Placeholder for future use)
eng-infra	(Placeholder for future use, this is currently called petname)
eng-security	(Placeholder for future use)

eng-support	(Placeholder for future use)	
sandbox	Descriptive name of the purpose (Ex. learning-kubernetes, tanuki-industries-iac-demo)	

Automated Infrastructure Environment ID (glenvid)

```
{{% alert color="info" %}}
This label/tag is optional.
{{% /alert %}}
```

```
terraform
glenvid: 'a1b2c3d4'
```

The environment ID allows various inputs, but in general is a short UUID, long UUID, SHA or other alphanumeric identifier that is systematically generated. The use of this label/tag will vary based on the glrealm that it is used in, and each team can use it however it makes sense to them.

Realm Usage Guidelines

Realm	Usage Guidelines or Standards	
-----	-----	
infra-shared-services	(Placeholder for future use)	
saas	(Placeholder for future use)	
business-tech	(Placeholder for future use)	
eng-dev	(Placeholder for future use)	
eng-infra	(Placeholder for future use)	
eng-security	(Placeholder for future use)	
eng-support	(Placeholder for future use)	
sandbox	For AWS accounts and GCP projects, this is the short UUID of the corresponding database record in the Sandbox Cloud portal that handled account provisioning. For Terraform-managed environments, this is the commit ID or SHA of the Terraform configuration (implementation varies).	

Geographic Continent (Region) of Environment (glenvcontinent)

```
{{% alert color="info" %}}
This label/tag is optional.
{{% /alert %}}
```

```
terraform
glenvcontinent: 'america'
```

> We use continent to avoid confusion with region since that is an ambiguous word.

This is the continent where this environment's resources are consumed by users. This may not necessarily be where it is deployed or where the environment owner resides. This allows us to understand impact to users if infrastructure maintenance is needed or security incidents occur.

This can also be used as meta data for global resources with autoscaling resources in specific continents/regions, however the glentity label/tag is needed for explicit cost allocation.

The abbreviation used by the business can change over time (Ex. APJ, APAC, Asia Pacific), and the cloud providers use different nomenclature, so we are standardizing using full names based on the prefix of programmatic timezones (Ex. America/LosAngeles is america). For non-regional services, you can use the global value.

Expected Values

Value	Business Acronyms	Cloud provider regions
global		(default) For global resources or any non-region specific services.
america	AMER, US, NORAM	AWS us-/ca-/sa-, GCP us-/southamerica-/northamerica-
europa	EMEA, METNA	AWS eu-/me-, GCP europe-
africa		AWS af-
asia	APAC, APJ	AWS ap-/cn-, GCP asia-
australia	APAC, APJ	AWS ap-, GCP australia-

> Some possible prefix values have been omitted from this list for brevity and unlikely applicability (Ex. brazil, hongkong, indian, etc.). They can be added at environment owner's discretion.

Owner Email Handle (glowneremailhandle)

```
{{% alert color="success" %}}
This label/tag is required.
{{% /alert %}}
```

```
terraform
glowneremailhandle: 'jmartin'
```

The email handle before @gitlab.com. We only use the handle since labels and tags do not support @ and . symbols. Values can be individual users or Google Workspace group aliases. For people handles, this should be {firstinitial}{lastname} and not any username aliases (Ex. jeff) since user aliases cannot be looked up programatically with our current architecture.

This is used for association with OKTA account, infrastructure notifications, and identifying the user's account with additional meta data. Users may receive emails with security advisory notifications.

Owner Timezone (glownertimezone)

```
{{% alert color="info" %}}  
This label/tag is optional.  
{{% /alert %}}
```

```
terraform  
glownertimezone: 'america-losangeles'
```

This is the timezone of the environment owner that is used for determining working hours. We assume that working hours are between 6am (06:00) and 6pm (18:00). This is usually only used when there is only one or two people who own a system. For systems with global coverage, you should not use this label/tag.

This is based on the programmatic timezone ID that usually follows the format Continent/Locality. You can see the full list of examples in the GCP documentation.

Due to limitations with the allowed characters in labels, the / is not allowed and we use lowercase labels. Therefore, we replace the slash / with a hyphen - and use lowercase letters exclusively. This is the same reason that we chose not to use UTC values since we cannot use + symbols.

Example Values

- america-losangeles
- america-denver
- america-chicago
- america-newyork
- asia-singapore
- asia-tokyo
- australia-sydney
- africa-johannesburg
- europe-amsterdam
- europe-berlin
- europe-london
- europe-paris
- europe-vienna

> These are examples and not an exhaustive list. As part of our inclusive values, please know that all timezones are valued equally and this was created based on some of the localities on the team map with the largest number of team members per capita.

GitLab Entity (glentity)

```
{{% alert color="info" %}}  
This label/tag is optional.  
{{% /alert %}}
```

```
terraform  
glentity: 'allocate'
```

This allows us to allocate costs to the respective business entity in financial reports. If not set, we will assume allocate.

Expected Values

Value	Description
allocate	(global default) Allocate based on % of revenue
inc	GitLab Inc, United States of America
ltd	GitLab Ltd, United Kingdom
bv	GitLab BV, the Netherlands
gmbh	GitLab GmbH, Germany
pty	GitLab PTY Ltd, Australia
canada	(future use) GitLab Canada Corp., Canada
gk	(future use) GitLab GK, Japan

See the GitLab Mailing addresses for details about each entity.

Realm Usage Guidelines

Realm	Usage Guidelines or Standards
infra-shared-services	(Placeholder for future use)
saas	(Placeholder for future use)
business-tech	(Placeholder for future use)
eng-dev	(Placeholder for future use)
eng-infra	(Placeholder for future use)
eng-security	(Placeholder for future use)
eng-support	(Placeholder for future use)
sandbox	The GitLab entity that the team member is associated with in Workday.

Workday Mapping

During testing of OKTA integration, this field is populating additional values we did not expect (ex. safeguard-italy, federal). This is documented in [www-data/data/entitymapper.yml](#) and we will map the respective entities to our cost center entities listed.

GitLab Department (gldept)

```
{{% alert color="success" %}}
This label/tag is required.
{{% /alert %}}
```

```
terraform
gldept: eng-dev
```

We have abbreviated all values to allow for easier prefixing using industry recognizable abbreviations. We use full words where an abbreviation is not universally recognizable. For

OKTA integration, we use YAML data file to remap values from Workday to their short names that we use for infrastructure. You can see a quick reference in the table below.

Expected Values

Division	Department	gldept (Slug/Label/Tag)
Engineering	Customer Support	eng-support
Engineering	Development	eng-dev
Engineering	Infrastructure	eng-infra
Engineering	Quality	eng-quality
Engineering	Security	eng-security
Engineering	UX	eng-ux
GA	Accounting	ga-accounting
GA	Business Technology	ga-business-tech
GA	CEO	ga-ceo
GA	Finance	ga-finance
GA	Legal	ga-legal
GA	People Success	ga-people
GA	Talent Acquisition	ga-talent-acquisition
Marketing	Awareness	mktg-awareness
Marketing	Brand & Digital Design	mktg-brand-design
Marketing	Campaigns	mktg-campaigns
Marketing	Communications	mktg-communications
Marketing	Developer Relations	mktg-community
Marketing	Content Marketing	mktg-content
Marketing	Digital Marketing	mktg-digital
Marketing	Field Marketing	mktg-field
Marketing	Inbound Marketing	mktg-inbound
Marketing	Marketing Ops	mktg-ops
Marketing	Owned Events	mktg-events
Marketing	Partner Marketing	mktg-partner
Marketing	Product Marketing	mktg-product
Marketing	Sales Development	mktg-sales-dev
Marketing	Strategic Marketing	mktg-strategic
Product	Product Management	product-mgmt
Sales	Business Development	sales-alliances
Sales	Channel	sales-channel
Sales	Commercial Sales	sales-commercial
Sales	Customer Success	sales-cs
Sales	Enterprise Sales	sales-ent
Sales	Field Operations	sales-field-ops
Sales	Practice Management	sales-practice-mgmt

GitLab Product Stage (glproductstage)

```
{{% alert color="info" %}}
```

This label/tag is optional.

```
{{% /alert %}}
```

```
terraform
glproductstage: eng-dev-manage
```

Since the eng-dev department has many groups, we use the Product sections, stages, groups and categories handbook page to define a parent group using the product stage. During the design discussion, we iterated with using product categories and sub-departments, and the stage made the most sense.

For GitLab SaaS and infrastructure cost allocation or attribution, the Engineering Infrastructure department team members can use the glproductstage label to attach to resources that should be attributed to a specific stage. You can optionally use the gldeptgroup if you need more granular attribution.

Expected Values

Value Product Category Documentation	
----- -----	

eng-dev-manage	Manage Stage
eng-dev-plan	Plan Stage
eng-dev-create	Create Stage
eng-dev-ecosystem	Ecosystem Stage
eng-dev-verify	Verify Stage
eng-dev-package	Package Stage
eng-dev-deploy	Deploy Stage
eng-dev-monitor	Monitor Stage
eng-dev-secure	Secure Stage
eng-dev-govern	Software Supply Chain Security Stage
eng-dev-growth	Growth Stage
eng-dev-fulfillment	Fulfillment Stage
eng-dev-enablement	Enablement Stage
eng-dev-modelops	ModelOps Stage
eng-dev-mobile	Mobile Stage
eng-dev-deploy	Deploy Stage

GitLab Product Category (glproductcategory)

```
{{% alert color="success" %}}
This label/tag is required.
{{% /alert %}}
```

```
terraform
glproductcategory: gitaly
```

Expected Values

Product categories should match one of the categories defined in categories.yml in the handbook: <<https://gitlab.com/gitlab-com/www-gitlab-com/-/blob/master/data/categories.yml>>

By providing this value, costs can be directly mapped back to a specific set of features, as well as a set of owners within both Product and Engineering.

GitLab Department Group (gldeptgroup)

{{% alert color="success" %}}

This label/tag is required.

{{% /alert %}}

terraform

gldeptgroup: eng-support-americas

For Engineering Development (eng-dev) department groups, the product stage slug (glproductstage) is used as the prefix for the gldeptgroup value. For all other departments, the gldept slug is used as the prefix.

All departments or stages have a default shared-infra label/tag/AWS account/GCP project that can be used when a separately distinguished/functional group is not needed. For groups with heavy infrastructure usage, we use the group name as a suffix and provide an isolated AWS account/GCP project for group team members to use.

For the groups that provide a shared service to other departments at GitLab, we also have a shared-services group value that can be used for shared cost attributions. We use shared-infra for our own team's resources and shared-services for everyone else's resources that we manage.

You can get an editable version of this table in the spreadsheet that we used to design this schema, however recent changes may not be reflected and this handbook page is the SSOT.

Expected Values

{{% alert color="warning" %}}

If a group listed below does not have a group documentation link, it is safe to assume that a AWS account or GCP project has not been created for that group yet. Please follow the instructions for provisioning the group (TODO).

{{% /alert %}}

<!-- To add group documentation, please add a link to a new page Group Documentation -->

The full list of groups was last audited and updated on 2021-07-22.

Footnotes:

1. If a group was renamed or removed, it appears in the Deprecated Group Names table below. Any renamed groups have a footnote indicator.

1. This was added after the initial list was created.

1. This is a special group that may not directly align with a team. For eng-dev, this usually appears under Other Functionality.

| glrealm | gldept | glproductstage | gldeptgroup

| Group Documentation |

----- ----- ----- -----			
----- -----			
infra-shared-services	wg-infra-shared-services		wg-infra-
shared-services [^2]			
eng-dev eng-dev	eng-dev-manage		eng-dev-
manage-shared-infra			
eng-dev eng-dev	eng-dev-manage		eng-dev-
manage-access			
eng-dev eng-dev	eng-dev-manage		eng-dev-
manage-import			
eng-dev eng-dev	eng-dev-manage		eng-dev-
manage-optimize [^1]			
eng-dev eng-dev	eng-dev-plan		eng-dev-
plan-shared-infra			
eng-dev eng-dev	eng-dev-plan		eng-dev-
plan-project-mgmt			
eng-dev eng-dev	eng-dev-plan		eng-dev-
plan-product-planning [^1]			
eng-dev eng-dev	eng-dev-plan		eng-dev-
plan-certify			
eng-dev eng-dev	eng-dev-create		eng-dev-
create-shared-infra			
eng-dev eng-dev	eng-dev-create		eng-dev-
create-source-code [^3]			
eng-dev eng-dev	eng-dev-create		eng-dev-
create-code-review [^2] [^3]			
eng-dev eng-dev	eng-dev-create		eng-dev-
create-editor			
eng-dev eng-dev	eng-dev-create		eng-dev-
create-gitaly			
eng-dev eng-dev	eng-dev-ecosystem		eng-dev-
ecosystem-integrations			
eng-dev eng-dev	eng-dev-ecosystem		eng-dev-
ecosystem-foundations			
eng-dev eng-dev	eng-dev-verify		eng-dev-
verify-shared-infra			
eng-dev eng-dev	eng-dev-verify		eng-dev-
verify-ci [^3]			
eng-dev eng-dev	eng-dev-verify		eng-dev-
verify-pipeline-authoring [^1] [^2]			
eng-dev eng-dev	eng-dev-verify		eng-dev-
verify-pipeline-execution [^1] [^2]			
eng-dev eng-dev	eng-dev-verify		eng-dev-
verify-runner			
eng-dev eng-dev	eng-dev-verify		eng-dev-
verify-testing			
eng-dev eng-dev	eng-dev-package		eng-dev-
package-shared-infra			
eng-dev eng-dev	eng-dev-package		eng-dev-

package-package			
eng-dev	eng-dev	eng-dev-deploy	eng-dev-
deploy-shared-infra			
eng-dev	eng-dev	eng-dev-deploy	eng-dev-
deploy-environments			
eng-dev	eng-dev	eng-dev-monitor	eng-dev-
monitor-shared-infra			
eng-dev	eng-dev	eng-dev-monitor	eng-dev-
monitor-monitor [^1] [^2]			
eng-dev	eng-dev	eng-dev-monitor	eng-dev-
monitor-apm [^3]			
eng-dev	eng-dev	eng-dev-monitor	eng-dev-
monitor-health [^3]			
eng-dev	eng-dev	eng-dev-secure	eng-dev-
secure-shared-infra			
eng-dev	eng-dev	eng-dev-secure	eng-dev-
secure-static-analysis			
eng-dev	eng-dev	eng-dev-secure	eng-dev-
secure-dynamic-analysis			
eng-dev	eng-dev	eng-dev-secure	eng-dev-
secure-composition-analysis			
eng-dev	eng-dev	eng-dev-secure	eng-dev-
secure-fuzz-testing			
eng-dev	eng-dev	eng-dev-secure	eng-dev-
secure-research			
eng-dev	eng-dev	eng-dev-govern	eng-dev-
govern-shared-infra			
eng-dev	eng-dev	eng-dev-govern	eng-dev-
govern-security-policies			
eng-dev	eng-dev	eng-dev-govern	eng-dev-
govern-threat-insights			
eng-dev	eng-dev	eng-dev-govern	eng-dev-
govern-compliance			
eng-dev	eng-dev	eng-dev-modelops	eng-dev-
modelops-shared-infra [^2]			
eng-dev	eng-dev	eng-dev-modelops	eng-dev-
modelops-applied-ml [^2]			
eng-dev	eng-dev	eng-dev-modelops	eng-dev-
modelops-ml-ops [^2]			
eng-dev	eng-dev	eng-dev-modelops	eng-dev-
modelops-data-ops [^2]			
eng-dev	eng-dev	eng-dev-growth	eng-dev-
growth-shared-infra			
eng-dev	eng-dev	eng-dev-growth	eng-dev-
growth-activation [^1] [^2]			
eng-dev	eng-dev	eng-dev-growth	eng-dev-
growth-adoption [^1] [^2]			
eng-dev	eng-dev	eng-dev-growth	eng-dev-
growth-conversion			
eng-dev	eng-dev	eng-dev-growth	eng-dev-

growth-expansion			
eng-dev	eng-dev	eng-dev-growth	eng-dev-
growth-retention			
eng-dev	eng-dev	eng-dev-growth	eng-dev-
growth-product-intelligence [^1]			
eng-dev	eng-dev	eng-dev-fulfillment	eng-dev-
fulfillment-shared-infra [^2]			
eng-dev	eng-dev	eng-dev-fulfillment	eng-dev-
fulfillment-shared-services [^2]			
eng-dev	eng-dev	eng-dev-fulfillment	eng-dev-
fulfillment-purchase [^1] [^2]			
eng-dev	eng-dev	eng-dev-fulfillment	eng-dev-
fulfillment-license [^1] [^2]			
eng-dev	eng-dev	eng-dev-fulfillment	eng-dev-
fulfillment-utilization [^1] [^2]			
eng-dev	eng-dev	eng-dev-enablement	eng-dev-
enablement-shared-infra			
eng-dev	eng-dev	eng-dev-enablement	eng-dev-
enablement-distribution			
eng-dev	eng-dev	eng-dev-enablement	eng-dev-
enablement-geo			
eng-dev	eng-dev	eng-dev-enablement	eng-dev-
enablement-search			
eng-dev	eng-dev	eng-dev-enablement	eng-dev-
enablement-sharding [^2]			
eng-dev	eng-dev	eng-dev-enablement	eng-dev-
enablement-database			
eng-dev	eng-dev	eng-dev-mobile (S.E.G.)	eng-dev-
mobile-devops-apps [^2]			
eng-dev	eng-dev	eng-dev-deploy (S.E.G.)	eng-dev-
deploy-five-min-app [^2]			
eng-infra	eng-infra		eng-infra-
shared-infra			
eng-infra	eng-infra		eng-infra-
shared-services			
eng-infra	eng-infra		eng-infra-
analytics			
eng-infra	eng-infra		eng-infra-
tenant-scale			
eng-infra	eng-infra	eng-infra-reliability	eng-infra-
reliability-shared-infra [^2]			
eng-infra	eng-infra	eng-infra-delivery	eng-infra-
automation [^2]			
eng-infra	eng-infra	eng-infra-scalability	eng-infra-
scalability			
eng-infra	eng-infra		eng-infra-
deliverability			
eng-infra	eng-infra	eng-infra-reliability	eng-infra-
reliability-database [^2]			
eng-infra	eng-infra	eng-infra-reliability	eng-infra-

reliability-foundations [^2]			
eng-infra eng-infra	eng-infra-reliability	eng-infra-	
reliability-general [^2]			
eng-infra eng-infra	eng-infra-reliability	eng-infra-	
reliability-observability [^2]			
eng-infra eng-infra	eng-infra-reliability	eng-infra-	
reliability-practices [^2]			
eng-infra eng-infra	eng-infra-enablement	eng-infra-	
cloud-connector			
eng-infra eng-infra	eng-infra-dev-ex	eng-infra-	
dev-ex-feature-readiness [^2]			
eng-dev eng-quality		eng-quality-	
shared-infra			
eng-dev eng-quality		eng-quality-	
env-toolkit [^2]			
eng-dev eng-quality		eng-quality-	
ops-ci-cd			
eng-dev eng-quality		eng-quality-	
secure-enablement			
eng-dev eng-quality		eng-quality-	
productivity			
eng-dev eng-quality		eng-quality-	
growth-govern			
eng-security eng-security		eng-	
security-shared-infra			
eng-security eng-security		eng-	
security-shared-services			
eng-security eng-security		eng-	
security-ops-infra			
eng-security eng-security		eng-	
security-ops-red			
eng-security eng-security		eng-	
security-ops-incident-response			
eng-security eng-security		eng-	
security-ops-trust-safety			
eng-security eng-security		eng-	
security-risk-compliance			
eng-security eng-security		eng-	
security-eng-app-sec			
eng-security eng-security		eng-	
security-eng-automation			
eng-security eng-security		eng-	
security-eng-research			
eng-security eng-security		eng-	
security-eng-vuln-mgmt			
eng-security eng-security		eng-	
security-logging			
eng-support eng-support		eng-support-	
shared-infra			
eng-support eng-support		eng-support-	

saas			
eng-support	eng-support		eng-support-
self-managed			
eng-support	eng-support		eng-support-
americas			
eng-support	eng-support		eng-support-
emea			
eng-support	eng-support		eng-support-
apac			
eng-dev	eng-ux		eng-ux-
shared-infra			
eng-dev	eng-ux		eng-ux-
technical-writing			
eng-dev	eng-ux		eng-ux-
research			
business-tech	ga-accounting		ga-
accounting-shared-infra			
business-tech	ga-business-tech		ga-business-
tech-shared-infra			
business-tech	ga-business-tech		ga-business-
tech-shared-services			
business-tech	ga-business-tech		ga-business-
tech-engineering	[^2]		
business-tech	ga-business-tech		ga-business-
tech-integrations	[^2]		
business-tech	ga-business-tech		ga-business-
tech-operations			
business-tech	ga-business-tech		ga-business-
tech-data			
business-tech	ga-business-tech		ga-business-
tech-procurement	[^2]		
business-tech	ga-ceo		ga-ceo-
shared-infra			
business-tech	ga-finance		ga-finance-
shared-infra			
business-tech	ga-legal		ga-legal-
shared-infra			
business-tech	ga-people		ga-people-
shared-infra			
business-tech	ga-recruiting		ga-
recruiting-shared-infra			
business-tech	mktg-awareness		mktg-
awareness-shared-infra			
business-tech	mktg-brand-design		mktg-brand-
design-shared-infra			
business-tech	mktg-campaigns		mktg-
campaigns-shared-infra			
business-tech	mktg-communications		mktg-
communications-shared-infra			
business-tech	mktg-community		mktg-

community-shared-infra			
business-tech	mktg-content		mktg-
content-shared-infra			
business-tech	mktg-digital		mktg-
digital-shared-infra			
business-tech	mktg-field		mktg-field-
shared-infra			
business-tech	mktg-inbound		mktg-
inbound-shared-infra			
business-tech	mktg-ops		mktg-ops-
shared-infra			
business-tech	mktg-events		mktg-
events-shared-infra			
business-tech	mktg-partner		mktg-
partner-shared-infra			
business-tech	mktg-product		mktg-
product-shared-infra [^2]			
business-tech	mktg-sales-dev		mktg-sales-
dev-shared-infra			
business-tech	mktg-strategic		mktg-
strategic-shared-infra			
business-tech	mktg-strategic		mktg-
strategic-technical-mktg			
eng-dev	product-mgmt		product-
mgmt-shared-infra			
business-tech	sales-alliances		sales-
alliances-shared-infra			
business-tech [^1]	sales-channel		sales-
channel-shared-infra			
business-tech	sales-commercial		sales-
commercial-shared-infra			
business-tech [^1]	sales-cs		sales-cs-
shared-infra			
business-tech [^1]	sales-cs		sales-cs-
demo-cloud			
business-tech [^1]	sales-cs		sales-cs-
training-cloud			
business-tech [^1]	sales-cs		sales-cs-
sa-shared-infra			
business-tech [^1]	sales-cs		sales-cs-
sa-us-west			
business-tech [^1]	sales-cs		sales-cs-
sa-us-east			
business-tech [^1]	sales-cs		sales-cs-
sa-pub-sec			
business-tech [^1]	sales-cs		sales-cs-
sa-mid-market			
business-tech [^1]	sales-cs		sales-cs-
sa-emea			
business-tech [^1]	sales-cs		sales-cs-

sa-apac			
business-tech [^1]	sales-cs		sales-cs-
tam-shared-infra			
business-tech [^1]	sales-cs		sales-cs-
tam-us-west			
business-tech [^1]	sales-cs		sales-cs-
tam-us-east			
business-tech [^1]	sales-cs		sales-cs-
tam-pub-sec			
business-tech [^1]	sales-cs		sales-cs-
tam-emea			
business-tech [^1]	sales-cs		sales-cs-
tam-apac			
business-tech [^1]	sales-cs		sales-cs-
ps-shared-infra			
business-tech [^1]	sales-cs		sales-cs-
ps-consulting			
business-tech [^1]	sales-cs		sales-cs-
ps-education			
business-tech [^1]	sales-cs		sales-cs-
ps-pub-sec			
business-tech	sales-ent		sales-ent-
shared-infra			
business-tech	sales-field-ops		sales-
field-ops-shared-infra			
business-tech	sales-practice-mgmt		sales-
practice-mgmt-shared-infra			

Deprecated Group Names

The following group names have been renamed or removed and may still be in use with legacy infrastructure and are preserved here for reference purposes.

Changelog Date	glrealm	gldept	glproductstage
gldeptgroup	Replaced with		
-----	-----	-----	-----
--- -----	-----		
2021-07-22	eng-dev	eng-dev	eng-dev-manage
eng-dev-manage-analytics		eng-dev-manage-optimize	
2021-07-22	eng-dev	eng-dev	eng-dev-plan
eng-dev-plan-portfolio-mgmt		eng-dev-plan-product-planning	
2021-07-22	eng-dev	eng-dev	eng-dev-create
eng-dev-create-gitter		N/A	
2021-07-22	eng-dev	eng-dev	eng-dev-release
eng-dev-release-progressive-delivery	N/A		
2021-07-22	eng-dev	eng-dev	eng-dev-growth
eng-dev-growth-acquisition		eng-dev-growth-activation, eng-dev-growth-adoption	
2021-07-22	eng-dev	eng-dev	eng-dev-growth
eng-dev-growth-telemetry		eng-dev-growth-product-intelligence	
2021-07-22	eng-dev	eng-dev	eng-dev-growth

eng-dev-growth-fulfillment	eng-dev-fulfillment-purchase, eng-dev-fulfillment-
license, eng-dev-fulfillment-utilization	
2021-07-22 business-tech ga-business-tech	
ga-business-tech-systems ga-business-tech-engineering, ga-business-tech-	
integrations	
2023-03-14 eng-dev eng-dev eng-dev-release	
eng-dev-release-shared-infra eng-dev-deploy-shared-infra	
2023-03-14 eng-dev eng-dev eng-dev-release	
eng-dev-release-release-mgmt eng-dev-deploy-environments	
2023-03-14 eng-dev eng-dev eng-dev-configure	
eng-dev-configure-shared-infra eng-dev-deploy-shared-infra	
2023-03-14 eng-dev eng-dev eng-dev-configure	
eng-dev-configure-configure eng-dev-deploy-environments	
2023-10-19 eng-dev eng-dev eng-dev-enablement	
eng-dev-enablement-memory eng-infra-cloud-connector	

Resource Type (glresourcetype)

```
{{% alert color="info" %}}
This label/tag is optional.
{{% /alert %}}
```

terraform

glresourcetype: compute-instance

GCP doesn't support easy reporting based on type of resource and name unless you use labels. This is optional, but recommended for resources that cost a significant amount of money or are part of a production service. This is not recommended for sandbox and ephemeral use cases.

Expected Values

Value	Recommended Usage
compute-cluster	AWS EKS, GCP Kubernetes Engine (GKE) cluster
compute-container	AWS ECS
compute-instance	AWS EC2 instance, GCP Compute Engine instance
serverless	AWS Lambda, GCP AppEngine
storage-disk	Persistent Storage Disks for instances
storage-snapshot	(Placeholder for discretionary usage, not easy to auto label/tag)
storage-bucket	AWS S3, GCP Google Cloud Storage (GCS)
network	AWS VPC, GCP Network/VPC

	network-lb	AWS ELB
	network-nat	AWS NAT
	network-firewall	Firewall rules
	network-ip	External IP addresses
	network-bandwidth	(Placeholder for discretionary usage)
	database	AWS RDS, AWS Aurora, GCP CloudSQL, etc.
	queue	AWS ElastiCache, MQ, SQS, GCP CloudTasks, etc.
	logging	(Placeholder for discretionary usage)

Resource Name (glresourcegroup)

```
{{% alert color="info" %}}
```

This label/tag is optional.

```
{{% /alert %}}
```

```
terraform
glresourcegroup: prod-app
```

This is optional for using with Terraform, Ansible, and other related infrastructure-as-code or related tools.

The value of this key should match the name of the group in your infrastructure-as-code group definitions (ex. Ansible role/group naming)

Resource Name (glresourcegroup)

```
{{% alert color="info" %}}
```

This label/tag is optional.

```
{{% /alert %}}
```

```
terraform
glresourcegroup: prod-app-web1
```

The value of this key should match the name of the resource in the cloud console.

GCP doesn't support easy reporting based on type of resource and name unless you use labels. This is optional, but recommended for resources that cost a significant amount of money or are part of a production service. This is not recommended for sandbox and ephemeral use cases.

Resource Host (glresourcehost)

{{% alert color="info" %}}

This label/tag is optional.

{{% /alert %}}

terraform

glresourcehost: prod-app-web1

This is used for associating child resources (disks, snapshots, buckets, network IPs, etc) with the parent resource (ex. compute-instance) to show the aggregate costs of the resource.

The value of this key should match the name of the parent resource in the cloud console.

GCP doesn't support easy reporting based on type of resource and name unless you use labels. This is optional, but recommended for resources that cost a significant amount of money or are part of a production service. This is not recommended for sandbox and ephemeral use cases.

Data Classification (gldataclassification)

{{% alert color="info" %}}

This label/tag is required for resources that store data (databases, storage buckets, etc..).

{{% /alert %}}

terraform

gldataclassification: red

Values should match the documented Data Classification Levels with all lowercase value (for cloud provider tag and label consistency).

Expected Values

Value	

red	
orange	
yellow	
green	

Accounting and Financial Reporting

In alignment with our transparency value, financial information cannot be shared publicly. Team members can see the details of how labels and tags impact financial reporting in this issue. The cost allocation methodology for production environments that in this document.

Impact to Business Owners and Infrastructure Team

The largest impact of the label and tag schema is known owner and department identification for

every resource provisioned in each cloud provider, and cross-reference to the Terraform configuration that can be referenced during infra/security review.

There are significant cost savings possible with the ability to set time limits on resources that solve the abandoned infrastructure problem. We can also save up to 65% of spending by implementing automation to power off infrastructure during non-working hours (overnight and weekends).

Impact to Users and Managers

With the label and tags, we can easily report on infrastructure and help you understand what's running and help you self manage the costs of the resources that you provision.

Since most labels and tags will be automatically added with Terraform and automation, or at the account/project level for sandbox accounts, there is not a significant burden on users to remember to add labels/tags to resources.

With future automation, we can also create frictionless experience for users with notification of system vulnerabilities, security incidents, and friendly reminders in Slack about upcoming expiration of infrastructure since your email, GitLab username, and Slack ID are automatically associated.

Footnotes

[^1]: This was renamed or removed after the initial list was created.

[^2]: This group was added after the initial list was created.

^3]: This is a special group that may not directly align with a team. For eng-dev, this usually appears under [Other Functionality].

SECTION: company/infrastructure-standards/policies.md

Overview

This is a placeholder for policies that we create for infrastructure standards.

SECTION: company/infrastructure-standards/realms/_index.md

Overview

This is a list of realms that have customized infrastructure standards based on each realm's business and technical requirements.

Realms

- infra-shared-services
- it

- saas
- security
- sandbox

SECTION: company/infrastructure-standards/realms/eng-infra/_index.md

Quick links

- Global infrastructure standards
- Global labels and tags
- Infrastructure policies
- Infrastructure helpdesk

Overview

This realm is for the engineering infrastructure team to deploy shared and team-specific infrastructure resources that aren't appropriate for the saas realms.

{{% panel header="Future Iteration with Engineering Infrastructure Handbook Pages" header-bg="info" %}}

The Engineering Infrastructure Environments handbook page is the current SSOT for environments. As the WIP initiative to iterate on our company-wide infrastructure standards evolves, the Engineering Infrastructure pages will be refactored incrementally as the standards are documented, implemented, and changes to environments take place.

{{% /panel %}}

Access requests

To request access to a group, please see group access request tutorial.

> For email authenticity security reasons, only GitLab issues or Slack messages to owners or counterparts are allowed for infrastructure requests.

Realm Owners

Name	GitLab.com Handle	Group Role	Job Title
-----	-----	-----	-----

Dave Smith	dawsmith	Owner	Engineering Manager, Reliability Engineering
Gerardo (Gerir) Lopez Fernandez	glopezfernandez	Counterpart	Engineering Fellow, Infrastructure

Realm labels and tags

The global labels/tags and realm labels/tags should be applied to each resource.

Realm Groups

Each infrastructure group has a shared GCP project and/or AWS account for group members.

If a group has not been implemented yet, please contact the realm owner for assistance. After a group is implemented, a separate handbook page is created with usage documentation.

| Group Name (AWS Account/GCP Project Name) | Usage Documentation (Empty cells are not implemented yet) |

----- -----		

eng-infra-shared-infra	<!--Group Docs-->	
eng-infra-shared-services	<!--Group Docs-->	
eng-infra-analytics	<!--Group Docs-->	
eng-infra-scalability	<!--Group Docs-->	
eng-infra-deliverability	<!--Group Docs-->	
eng-infra-reliability	<!--Group Docs-->	

Usage guidelines

This is a placeholder for the realm owner to provide instructions on best practices and usage guidelines for this infrastructure.

SECTION: company/infrastructure-standards/realms/eng-infra/labels-tags.md

Quick Links

- Infrastructure Standards - Sandbox Realm Documentation
- Infrastructure Standards - Global Labels and Tags Standards
- Infrastructure Standards - Policies
- Infrastructure Standards - Tutorials
- Infrastructure Standards - Helpdesk

Quick Reference

We use the gl prefix for all labels and tags. All keys use underscores (snakecase). All values should use hyphens (alpha-dash for slug'd values), however underscores are allowed.

Global Labels/Tags

See the list of global labels and tags that should be applied to each resource.

Realm Labels/Tags

There are no labels and tags specific to this realm yet.

SECTION: company/infrastructure-standards/realms/infra-shared-services/_index.md

Quick links

- Global infrastructure standards
- Global labels and tags
- Infrastructure policies
- Infrastructure helpdesk

Overview

This is a placeholder page for the infra-shared-services realm.

SECTION: company/infrastructure-standards/realms/infra-shared-services/labels-tags.md

Quick Links

- Infrastructure Standards - Sandbox Realm Documentation
- Infrastructure Standards - Global Labels and Tags Standards
- Infrastructure Standards - Policies
- Infrastructure Standards - Tutorials
- Infrastructure Standards - Helpdesk

Quick Reference

We use the gl prefix for all labels and tags. All keys use underscores (snakecase). All values should use hyphens (alpha-dash for slug'd values), however underscores are allowed.

Global Labels/Tags

See the list of global labels and tags that should be applied to each resource.

Realm Labels/Tags

There are no labels and tags specific to this realm yet.

SECTION: company/infrastructure-standards/realms/it/_index.md

Quick links

- Global infrastructure standards
- Global labels and tags
- Infrastructure policies
- Infrastructure helpdesk

Overview

This infrastructure realm is for resources managed by the IT Infrastructure team.

Access requests

To request access to a group, please see [group access request tutorial](#).

> For email authenticity security reasons, only GitLab issues or Slack messages to owners or counterparts are allowed for infrastructure requests.

Realm Owners

Name	GitLab.com Handle	Group Role	Job Title
-----	-----	-----	-----

Jeff Martin	jeffersonmartin	Owner	Sr IT Systems Engineer
Dillon Wheeler	dillonwheeler	Counterpart	IT Systems Engineer
Mohammed Al Kobaisy	malkobaisy	Counterpart	IT System Administrator
Daniel Parker	dparker	Counterpart	Sr Integrations Engineer

Realm labels and tags

The global labels/tags and realm labels/tags should be applied to each resource.

Usage guidelines

This is a placeholder for the realm owner to provide instructions on best practices and usage guidelines for this infrastructure.

SECTION: company/infrastructure-standards/realms/it/labels-tags.md

Quick Links

- Infrastructure Standards - Sandbox Realm Documentation
- Infrastructure Standards - Global Labels and Tags Standards
- Infrastructure Standards - Policies
- Infrastructure Standards - Tutorials
- Infrastructure Standards - Helpdesk

Quick Reference

We use the `gl` prefix for all labels and tags. All keys use underscores (snakecase). All values should use hyphens (alpha-dash for slug'd values), however underscores are allowed.

Global Labels/Tags

See the list of global labels and tags that should be applied to each resource.

Realm Labels/Tags

There are no labels and tags specific to this realm yet.

SECTION: company/infrastructure-standards/realms/saas/_index.md

Quick links

- Global infrastructure standards
- Global labels and tags
- Infrastructure policies
- Infrastructure helpdesk

Overview

This is a placeholder page for the saas realm.

{{% panel header="Future Iteration with Engineering Infrastructure Handbook Pages" header-bg="info" %}}

The Engineering Infrastructure Environments handbook page is the current SSOT for environments. As the WIP initiative to iterate on our company-wide infrastructure standards evolves, the Engineering Infrastructure pages will be refactored incrementally as the standards are documented, implemented, and changes to environments take place.

{{% /panel %}}

SECTION: company/infrastructure-standards/realms/saas/labels-tags.md

Quick Links

- Infrastructure Standards - Sandbox Realm Documentation
- Infrastructure Standards - Global Labels and Tags Standards
- Infrastructure Standards - Policies
- Infrastructure Standards - Tutorials
- Infrastructure Standards - Helpdesk

Quick Reference

We use the gl prefix for all labels and tags. All keys use underscores (snakecase). All values should use hyphens (alpha-dash for slug'd values), however underscores are allowed.

Global Labels/Tags

See the list of global labels and tags that should be applied to each resource.

Realm Labels/Tags

There are no labels and tags specific to this realm yet.

SECTION: company/infrastructure-standards/realms/sandbox/_index.md

Quick links

User Portal

- gitlabsandbox.cloud

Documentation

- Global infrastructure standards
- Global labels and tags
- Infrastructure policies

Issue Tracking and Collaboration

- HackyStack issue tracking (open source code feature development)
- CorpSec Infra issue tracking (GitLab-specific topics and requests)
- sandbox-cloud-questions Slack channel to ask questions and get help.

Code and Examples

- HackyStack README
- HackyStack source code
- HackyStack screenshots

Infrastructure-as-Code

- Sandbox Cloud - Project Templates
- Sandbox Cloud - Terraform Modules
- Sandbox Cloud - Ansible Roles
- IT Infrastructure Realm IaC - Terraform Configuration for gitlabsandbox.cloud - Restricted access
- IT Infrastructure Realm IaC - Ansible Configuration for gitlabsandbox.cloud - Restricted access

Overview

The Sandbox Cloud is an automated provisioning platform for creating an AWS account or GCP project that you can use for demo/sandbox/testing purposes and is paid for by GitLab with consolidated invoice billing (no credit card required).

This platform is powered by HackyStack, an open source project created by Jeff Martin and maintained by Jeff Martin and Dillon Wheeler to automate the access request process using Okta integration, auto-assigning roles and permissions based on your department, and using the cloud provider API for provisioning your AWS account and/or GCP project.

You can learn more in the HackyStack High-Level Intro presentation.

The Sandbox Cloud is managed by the IT Engineering team. Please tag Jeff Martin in Slack with any questions.

How to Get Started

> We are currently at beta stability. Please post in sandbox-cloud-questions if you see unexpected behavior.

Individual AWS Account or GCP Project

Any team member can use the self service instructions below to provision an AWS account or GCP project for your individual use (sandbox, testing, etc.). You cannot invite other team members to individual accounts for security reasons.

1. Visit <https://gitlabsandbox.cloud> and sign in with your Okta account.
1. Navigate to Cloud Infrastructure in the top navigation.
1. Click the purple Create Individual Account button.
1. Choose the cloud provider and cloud organization unit from the dropdown menu. If no options are present in the dropdown list for the organization unit, your department has not been created in our database yet due to a department name change or addition in the HRIS. Please ask in sandbox-cloud-questions to have it added.
1. Click the green Create Account button.
1. Your account will take 2-5 minutes for the AWS API to finish the provisioning process while the AWS services are activated for your account.
1. Please refresh your browser window every ~60 seconds until you see that your user account has changed from Provisioning to Active.
1. See the instructions below for Accessing your AWS Account or Accessing your GCP Project.

> You can sign-in with Okta, however please don't create a Cloud Account unless you intend to provision AWS resources. You can see the screenshots of everything that a user sees.

Is your current AWS account experience problems? Please ask for help in sandbox-cloud-questions. If your problems are validated and approved for getting a new AWS account, please use the New AWS Individual Account Rebuild Request issue template.

Automated Shutdown Policy

For cost saving and exposure reduction measure, any GCP compute engine instances in an individual account (ex. dmurphy-a1b2c3d4) are powered down every Friday at 23:59 UTC starting 2023-02-03. You will receive a Slack bot notification 24 hours in advance for any instances that are affected with instructions for adding a label to the instance if you need to prevent power off.

If an instance is powered off, you can simply power the instance back on when you're ready to work again.

We are not charged for the hours that compute instances that are not powered on (e2-standard-4/4vCPU/16GB costs \$0.13/hr or \$96.48/mo). Storage for a powered down instance is cheap and a 20GB persistent disks cost \$0.80/month.

This has immediate 28%+ cost savings by not having ephemeral infrastructure running on Saturdays and Sundays. Additional cost savings since we are not charged until you power the instance back on (could be days, weeks, or months later). This also covers abandoned instances that were only used for a few hours for a demo.

We will be working on AWS accounts in a future iteration.

Group/Team AWS Account or GCP Project (Non-Production)

Any team member can request a new AWS account or GCP project for a specific project or working group, or for general usage by their department group for shared non-production resources.

No RED data is allowed in these accounts/projects. Any RED data must be hosted in production AWS accounts or GCP projects managed by the appropriate Infrastructure Realm administrators (ex. eng-infra-saas, it-infra, etc.).

Self-service creation and IAM management is not available yet for end users in HackyStack. In the meantime, we use access request style issue templates as our boring solution for security compliance reasons and the HackyStack administrators provision accounts and users using the Admin CLI.

- Issue Template: New AWS Group Workload (Multi-user) Account Request
- Issue Template: Add/Remove IAM Users from AWS Group Workload Account
- Issue Template: New GCP Group (Multi-user) Project Request (Provisioner Runbook)
- Issue Template: Add/Remove IAM Users from GCP Group Project

Production Environments

Product Related

For any staging or production(-esque) infrastructure services that are customer facing, contain Red or Orange data, related to the GitLab product or GitLab.com SaaS, or Engineering sponsored services, please contact the Reliability Engineering team for guidance on next steps in the infrastructurelounge Slack channel.

Most environments are typically created in the config-mgmt project using the Create a new environment instructions.

You can learn more about GitLab.com SaaS on the Production Architecture handbook page.

Any projects with yellow or green data usually are better suited for self management using Group Projects using Infrastructure Standards guidelines.

Business Related

For any infrastructure services related to business operations and our tech stack, please contact the IT team in ithelp for guidance on next steps. Most of our tech stack are SaaS-based and hosted by the respective vendor.

New SaaS applications should go through the Procurement Process and are managed by the respective department's system owners.

Self-hosted application infrastructure is determined on a case-by-case basis and is architected in collaboration with CorpSec Infrastructure, Infrastructure Security, Application Security, and 3rd Party Risk. Please tag @vlad in an issue for preliminary guidance on new services. If you do not have an issue yet, please create one in the CorpSec Infrastructure issue tracker.

Accessing your AWS Account

1. Sign in to <https://gitlabsandbox.cloud> and navigate to Cloud Infrastructure.
1. If you have not created an account yet, click the Create Individual Account button and use the form to create a new account. If you already have an account, click on the title or the gear icon for the AWS account you want to access.
1. On the Cloud Account details page, click the View IAM Credentials button in the top right corner to open up a popup modal window.
1. You will see the AWS Console URL, Username, and Password that you can use to sign in to your AWS account. The 12 digit number at the beginning of the URL is your AWS Account ID/Number.
1. Create a new 1Password record in your Private vault to save these credentials.
1. You can click on the link to open the AWS console, or you can close the modal window and click the Open AWS Web Console button on the Cloud Account details page.
1. Use the provided URL, Username, and Password to sign in to your new AWS account. Be careful that your browser doesn't autofill saved credentials for a different account.
1. After you sign in, you should navigate to IAM and add a Virtual MFA device for your user account and add a One-Time Password (OTP) to your 1Password record.
1. Your IAM user account has AdministratorAccess to be able to perform any action inside of your AWS account. We do not provide team members access to the root us